Arista Access Node MG-ACC-70 QoS Congestion Event ALARM_9981 Minor SOP

Node: MG-ACC-70 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_9981

Description

This Standard Operating Procedure (SOP) addresses the scenario where the Arista Access Node MG-ACC-70, situated in the MG Road Core, experiences a Quality of Service (QoS) congestion event, manifesting as microbursts, interface tail drops, or QoS queue starvation. This phenomenon is primarily attributed to the complex interplay of QoS policies, buffer sizes, and packet scheduling algorithms on the Arista Access Node. The QoS configuration, which is based on the IEEE 802.1p and DSCP (Differentiated Services Code Point) standards, enables the prioritization of traffic flows according to their designated class of service. However, if the buffer size is insufficient to handle the incoming traffic, or if the packet scheduling algorithm is misconfigured, it can lead to QoS queue starvation, causing the interface to drop packets and resulting in the ALARM_9981 alarm. Furthermore, the presence of microbursts can also contribute to the congestion event. These microbursts are short, intense bursts of packets that can overwhelm the interface, leading to packet drops and further exacerbating the congestion.

Pre-requisite Checks

- Verify that the MG Road Core site is accessible and that the remote hands are dispatched.

- Check the redundant link status to ensure that it is operational and not contributing to the congestion event.

- Confirm that the QoS configuration is up-to-date and aligned with the current network policies.

Potential Root Causes

Cause	Probability
Inadequate buffer size or misconfigured packet scheduling algorithm	High
Microbursts on the interface	Medium
QoS queue starvation due to incorrect QoS configuration	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware cpu usage, Expected Output: CPU usage information with high CPU utilization, Description: This command reveals the CPU utilization on the Arista Access Node, providing insight into whether the congestion event is CPU-bound or not.

Command: show interface counters, Expected Output: Interface counters with high packet drops and errors, Description: This command displays the interface counters, helping to identify if the congestion event is caused by packet drops or errors.
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```
Command: show qos policy-map, Expected Output: QoS policy-map configuration with incorrect class-map or policy-map configuration, Description: This command reveals the QoS policy-map configuration, enabling the identification of any misconfigured class-map or policy-map that may be contributing to the congestion event.
```

Resolution Steps

Step 1: Step: Increase the buffer size on the interface, Command: hw-module profile buffer-size , Description: This step involves increasing the buffer size on the interface to accommodate the incoming traffic and prevent QoS queue starvation.

Step 2: Step: Reconfigure the packet scheduling algorithm, Command: hw-module profile scheduler , Description: This step involves reconfiguring the packet scheduling algorithm to ensure that packets are scheduled correctly and not causing QoS queue starvation.

Step 3: Step: Mitigate microbursts on the interface, Command: hw-module profile microburst , Description: This step involves mitigating microbursts on the interface to prevent packet drops and further exacerbating the congestion event.

Post-Check Verification

```
Command: show qos policy-map, Expected Output: QoS policy-map configuration with corrected class-map or policy-map configuration, Description: This command ensures that the QoS policy-map configuration is corrected and not contributing to the congestion event.

Command: show interface counters, Expected Output: Interface counters with reduced packet drops and errors, Description: This command verifies that the packet drops and errors have been reduced, indicating that the resolution steps have been effective.

Command: show hardware cpu usage, Expected Output: CPU usage information with reduced CPU utilization, Description: This command confirms that the CPU utilization has been reduced, indicating that the congestion event has been mitigated.
```

Roll-back Procedures

- Step: Restore the original buffer size on the interface, Command: hw-module profile buffer-size , Description: This step involves restoring the original buffer size on the interface to its previous state in case the resolution steps fail.

- Step: Restore the original packet scheduling algorithm, Command: hw-module profile scheduler , Description: This step involves restoring the original packet scheduling algorithm to its previous state in case the resolution steps fail.

- Step: Restore the original microburst configuration, Command: hw-module profile microburst , Description: This step involves restoring the original microburst configuration to its previous state in case the resolution steps fail.